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# LAPLACIAN SHARPENING WITH DIAGONAL NEIGHBORS
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```
import cv2
import numpy as np
import matplotlib.pyplot as plt
from google.colab import files
```

```
# Upload Image
uploaded = files.upload()

filename = list(uploaded.keys())[0]
```

```
# Read Image
img = cv2.imread(filename)
img_rgb = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)
```

```
# Laplacian Kernel
kernel = np.array([
    [1, 1, 1],
    [1,-8, 1],
    [1, 1, 1]
])
```

```
# Apply Kernel
laplacian = cv2.filter2D(img_rgb, -1, kernel)
```

```
# Sharpened Image
sharpened = cv2.subtract(img_rgb, laplacian)
```

```
# Display
plt.figure(figsize=(12,5))
```

```
plt.subplot(1,2,1)
plt.imshow(img_rgb)
plt.title("Original Image")
plt.axis("off")
```

```
plt.subplot(1,2,2)
plt.imshow(sharpened)
plt.title("Sharpened Image")
plt.axis("off")
```

```
plt.show()
```

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Original Image



Sharpened Image

